

साक्षी

# Saakshi

A resilient, provable computer-based testing ecosystem

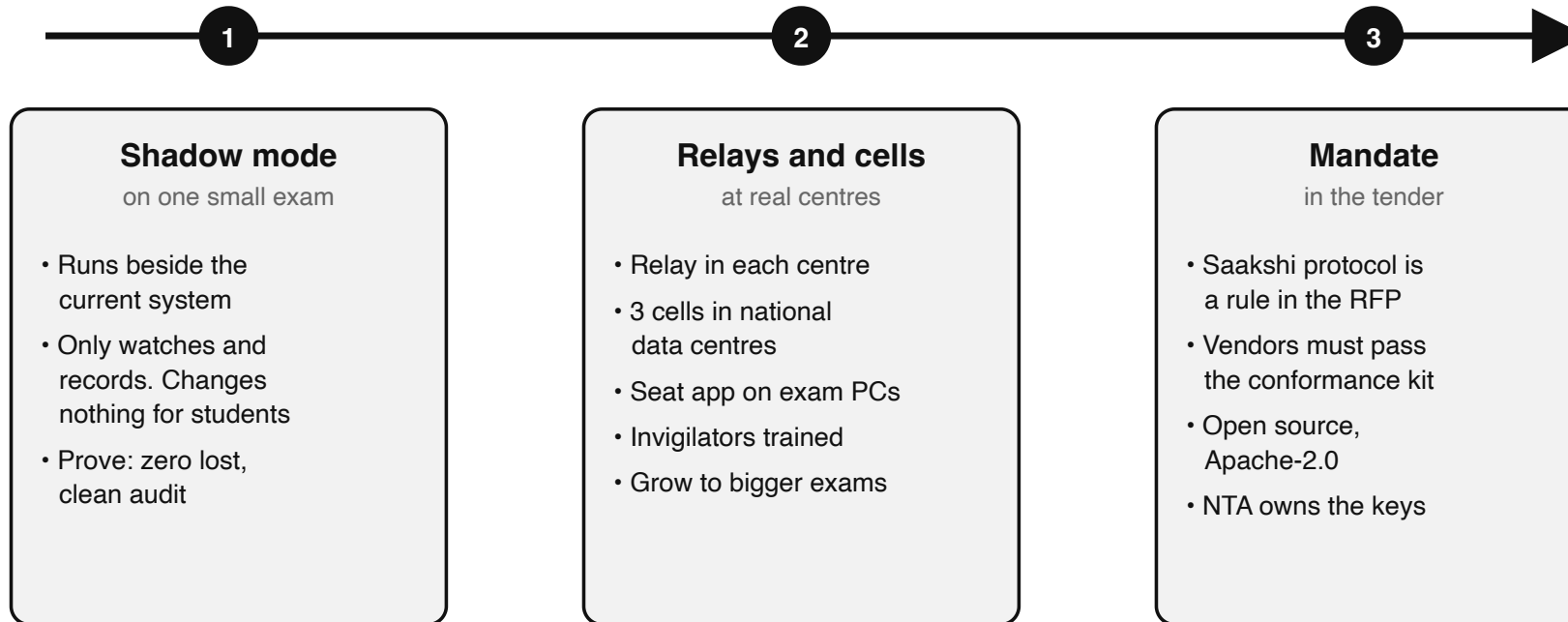
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Hackathon Challenge 6 · Implementation & Feasibility Plan

*Team Kawaai: Anirudh, Bhumika, Arjav*

# Can it really be built?

Short answer: the core already runs. The rest is rollout, in three steps.



**Why shadow mode first?** Nobody has to trust us on day one. The old system keeps running. We only compare.

# Who does what

| Who       | Policy and keys | Run control and cells | Run centres            | Watch from outside         |
|-----------|-----------------|-----------------------|------------------------|----------------------------|
| NTA       | Owns            | Approves              | Approves               | Gets reports               |
| NIC       | Advises         | Owns                  | -                      | -                          |
| Vendors   | -               | -                     | Own, must pass the kit | -                          |
| Witnesses | -               | -                     | -                      | Own, outside the exam path |

Witnesses sit outside the exam path, so they can check without being able to change anything.

# What already works

Measured on one Apple M5 Mac with 3 real cells and 99 simulated centres. Tested on macOS and Windows.

**20,000**

candidates in the load test

**227,882**

entries stored, 0 rejected

**20 / 20**

chaos runs passed

**0**

answers lost out of 16,000

**44 ms**

median ingest (p50)

**168 ms**

p99 ingest

**~1 s**

recovery, median 0.96 to 1.23 s

**1,000 / 1,000**

tampered records located

**Seat**  
one tick**Relay**  
two ticks**Cell**  
blue ticks**Audit**  
sealed**Decide**  
fair result

**Honest limits.** Everything ran on one machine over loopback, not a real WAN. Analytics numbers use synthetic data. Power-loss and 20k over a real network are not yet measured.

# Checklist of what is done

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- |                                                                               |                                                                                 |
|-------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| ■ Seat app signs and saves every answer, even offline                         | ■ Candidate moves to another seat and keeps the time they had                   |
| ■ Relay in the centre cannot read answers                                     | ■ Audit page shows exactly which record was edited                              |
| ■ Three cells verify and countersign; a wiped cell rebuilds from relays       | ■ Analytics flags likely cheating; no honest candidate flagged (synthetic data) |
| ■ Spare relay takes over during an exam                                       | ■ 476 workspace tests and 76 analytics tests pass                               |
| ■ Paper released only when 2 of 3 custodians agree; offline unlock code works |                                                                                 |

**Why it can be implemented.** It runs on today's centre PCs, uses standard cryptography (SHA-256, ECDSA P-256), and is open source under Apache-2.0 with a frozen protocol and fixed test vectors.

# What it needs

| Part           | What is needed                                                      | Evidence                 |
|----------------|---------------------------------------------------------------------|--------------------------|
| Exam PC (seat) | Windows 10 x64 or macOS 12, 4 GB RAM, dual core. Camera optional    | Minimum spec in plan     |
| Seat load      | 2.3% CPU median (max 4.3%), 166 MiB memory                          | Measured                 |
| Centre link    | About 1.06 MB per candidate per hour                                | Measured                 |
| Centre relay   | One small machine per centre, plus a spare running the same program | By design                |
| Cells          | 3 independent processes in national data centres                    | Ran 3 on one Mac         |
| Bigger exams   | About 10 cells for 1 million candidates at once                     | Assumption, not measured |

**Cost.** Saakshi is open source (Apache-2.0), so there is no licence fee. A full rupee cost per candidate has not been worked out yet. We will not quote a number we have not measured.

# Risks and fixes

| Risk                                             | What we do                                                                  |
|--------------------------------------------------|-----------------------------------------------------------------------------|
| Camera check fails on some PCs                   | Invigilator confirms identity by hand instead                               |
| Windows hardware arrives late                    | Packaged app is tested automatically on Windows; demo falls back to the Mac |
| Rules change and break old records               | Protocol v1 is frozen and checked with fixed test vectors                   |
| Speed claims do not hold up                      | Publish only measured numbers; say where they were measured                 |
| Fear of false cheating flags                     | Thresholds published before the exam; a human signs every decision          |
| Too much scope                                   | Shadow mode first; grow only after each step passes                         |
| Real network is slower than our test             | Seats keep working offline and sync later; early warning when sync lags     |
| One signature does not prove who chose an answer | Stated openly; identity check at entry and invigilator attestation          |